Utilizing Reinforcement Learning with Path Planning Algorithms to Create More Efficient Paths of Travel

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***Abstract*—In this report, the implementation of a reinforcement learning algorithm in conjunction with multiple basic path planning algorithms is investigated to determine if reinforcement learning can efficiently optimize a simulated TurtleBot in a 2D environment based on the manipulation of several key parameters with the performance measured against path length, smoothness, travel time, velocity, and clearance. The simulations under study were conducted with ROS2 in a Gymnasium environment using open-source Turtlebot code and a Nav2 stack. The reinforcement learning algorithm was designed to utilize MPPI and DWB path planners and as a result, optimize them to perform better than their current offline counterparts.**

I. Introduction

A. Current state of the Art

In the industrial environment of Autonomous Mobile Robots (AMRs), path planners are generally deployed en-masse and are not tailored to the customers or environment needs in which they are deployed. This results in good, but not excellent, performance and, while easier to maintain on the backend, can limit customer output on the front end. These general deployments often cast a wide net and aim to operate on the "safe side", whether that be wide, nonadjustable clearance, paths around high traffic areas, or creating lanes that avoid foot traffic altogether. In reality, each customer has different needs and, outside of required safety standards, may have certain workplace standards that could offer better performance if utilized properly to tune the AMR deployment. It is obvious that hand tuning any number of parameters for each individual deployment site is unreasonable and the time spent on it would far outweigh any long-term benefits. Additionally, the deployment time for standard AMRs can be several weeks, meaning there is a potential for downtime when looking at brownfield sites. Implementing a reinforcement learning (RL) algorithm paired with machine vision such as LiDAR could reduce that deployment time from weeks to days, as all the training is done on the back end and there is no need to manually set path layouts.

B. Project Objectives

To reduce the amount of manual input needed to tune path planner parameters, a reinforcement learning (RL) algorithm is deployed into the simulation for a period of training, during which the following parameters are adjusted to optimize the Turtlebot travel across the simulated warehouse: Clearance, Accuracy, Efficiency, and Smoothness. These four parameters have been previously proven to be statistically significant modifiers for both path planners under study [1] [2]. In this context, a dense reward function is custom-built to help direct the RL algorithm through each episode so that when it is deployed, it can be more lightweight in terms of computational load.

C. Background

While modern RL algorithms are a fairly new technology, AMRs and their predecessors, Automated Guided Vehicles (AGVs), are not and have been in use for over 70 years in various forms. Similarly, path planners such as Model Predictive Path Integral (MPPI) and Dynamic Window Approach-Based (DWB) are 10-15 years old. As a result, there is a wide breadth of research that has been conducted to both verify plausibility as well as serve as a reasonable set of starting points for this project.

D. Path planning

Elbouhy et. al designed a study that consisted of three different path planners that were tested in environments with static and dynamic obstacles and the results were then evaluated to compare algorithms. The study analyzed DWB, MPPI, and RPP (Improved Pure Pursuit) planners and used the following criteria to determine path quality: time, distance, smoothness, minimum clearance, runs completed, and number of recoveries. This study showed that MPPI was the fastest but least smooth, DWB operated with the least clearance, and RPP took the most time and traveled the longest distance [1]. This demonstrates that different planners have different strengths and weaknesses which ultimately led to the integration of both

MPPI and DWB into the RL-based simulation built in this project.

Li et. al studied the Dynamic Window Approach (DWA) algorithm and adjusted evaluation parameters in an effort to optimize the path planning. The evaluation function utilized in this paper looked at the heading, the distance between the trajectory and nearest obstacle, and the speed of the robot, along with weighted modifiers of each. A simple environment was setup and the tuning modifiers were adjusted for optimization [2]. In the context of this project, the interest lies in the tuning equation itself; and Li's tuning equation provided a confident proof-of-concept that path planning algorithms can be tuned and optimized on a limited number of parameters and the results are statistically significant and positive.

E. Reinforcement Learning

Qu et. al. presented a method of training an RL policy offline to accelerate the convergence for the online control of MPPI in an unmanned aerial vehicle (UAV). The method they proposed involved utilizing the Distributional Soft Actor-Critic (DSAC) algorithm to have an RL agent train a pair of neural networks to model distributions for the control costs and roll-out guiding policy used directly within the online MPPI control. This improved MPPI by providing better "first guesses" than standard MPPI. The results of their experiments showed that their method converged significantly faster, produced smoother trajectories, and provided higher success rate in reaching the goal than standard MPPI [3]. The guiding principle of this paper translates directly to this project to enhance the strengths of existing local planners instead of replacing them by leveraging the strengths of both RL and traditional path planning algorithms to create a more efficient and adaptable path planning solution for mobile robots in complex environments.

Ting et. al. presented a modified Proximal Policy Optimization (PPO) in conjunction with a deep reinforcement learning (DRL) algorithm in order to reduce the time needed to move an agent through an environment. To reduce the number of episodes required and thus the calculation time for each move, they proposed a distance-based weighted dense reward function. The reward function they proposed utilized the euclidean distance to the goal to provide a higher value reward for better steps [4]. Since this reward function is not strictly linked to PPO parameters, the concept can easily translate into both MPPI and DWB that are utilized for the training in this project.

Debnath et al. propose a modular navigation framework for navigation that combines a classical global planner with a DRL based local planner. The global planner generates waypoints and assigns Gaussian soft costs around detected pedestrians to anticipate their future positions, while the trained local planner dynamically follows these waypoints while avoiding imminent collisions in real time. The reward function used to train the local planner includes eight terms covering goal reaching, collision penalties, waypoint tracking progress, pedestrian proximity avoidance, and a small per-timestep penalty to encourage efficiency. Through experiments, their hybrid system outperformed both a traditional DWA planner and a standalone RL baseline, with the RL-augmented system achieving a notably lower collision rate [9]. This paper is directly relevant to the dynamic parameter tuning approach adopted in this project where the RL agent observes local state and adjusts planner parameters throughout the navigation process rather than making a single configuration decision upfront.

Rana and Kaveendran implemented a PPO-based DRL agent to train a TurtleBot3 navigation policy from scratch within a ROS2 environment. The agent observes the robot's surroundings through a LiDAR scan and directly outputs normalized linear and angular velocity commands at each timestep, making it the sole navigation controller rather than an augmentation of an existing planner. Training was conducted using a two-term reward function including a small positive reward for forward progress and a large negative penalty for collisions. The trained policy resulted in better performance, and behavior improvements such as slowing near tight spaces [10]. This paper is particularly relevant as a validation of a similar implementation being used in this project: ROS2, Gymnasium, PPO, and a TurtleBot platform. The key distinction is that their RL agent replaces the planner entirely by outputting direct velocity commands, whereas this project's agent acts as a meta-controller. Their training time was approximately four hours for a significantly harder problem (the agent must learn full navigation from raw sensor data), providing reassurance that the team's proposed approach is viable.

F. Other Noteworthy Resources

In addition to the papers described in detail above, there were several other papers that were reviewed which provided supplemental information for shaping the project. [5] and [6] provided metrics and insights on planner path evaluation which was utilized as inspiration for the evaluation metrics implemented in this project. [7] and [8] demonstrated more examples which utilized RL within some sort of path planning framework for a mobile robot (including both direct and indirect RL control), again reinforcing the possibility of the concept explored in this project.

While there are several papers which demonstrate the use of RL in path planning, no papers were identified which explicitly utilized RL to configure the parameters for a given local path planning algorithm. This is a critical gap in the literature and study of this field, as it is well established that the performance of local path planners can be highly sensitive to their parameter settings, and that optimal settings can vary significantly across different environments and use cases. Meanwhile, some of the examples utilizing direct RL control are not very extensible as these methods tend to only train the RL agent to operate in a specific environment; and re-training for more complex environments can be very costly. By leveraging RL to automatically tune these parameters, this project theoretically fills this gap and demonstrates a novel approach to optimizing path planning performance in a way

that is adaptable to the specific needs of different deployment sites.

II. Materials and Methods

A. Simulation Setup

In order to reduce the scope and complexity of the required programming of this project, an RL agent was implemented within ROS2. Using ROS2 prevents needing to develop a simulation environment and navigation stack from scratch; which allows for more dedicated effort to developing the RL infrastructure and evaluating performance. Additionally, the simulation environment is constrained to a 2D grid with a TurtleBot robot model in order to simplify the state space and holonomic constraints for the path planners and RL infrastructure, again to avoid adding unnecessary complexity.

The environment was designed to be cross-platform and reproducible, leveraging Docker to encapsulate all dependencies and ensure consistent behavior across the team's Mac, Windows, and Linux development machines. The environment is structured as a two-layer Docker image: a base capstone_ros_layer image that installs all system-level dependencies (ROS2, Nav2, TurtleBot4 description packages, and visualization tools), and a top-level rbe_capstone image that layers in the project's ROS2 workspace. Separating dependency installation from application code allows the more expensive base layer to be rebuilt infrequently while code iteration remains fast.

The simulation runs on ROS2 Jazzy with the CycloneDDS middleware layer for inter-node communication. The robot model used is the TurtleBot4, whose URDF is loaded via `xacro` and published to the `/robot_description` topic by the `robot_state_publisher` node. Navigation is performed in a 2D occupancy grid environment using the Nav2 stack, which includes `nav2_map_server` for loading prebuilt maps and `nav2_lifecycle_manager` for managing the life cycle of navigation nodes. Maps are stored as paired `.pgm` and `.yaml` files and are selectable at launch time, enabling the team to quickly swap between different warehouse environments as the project progresses.

Visualization is handled by RViz2, which renders the robot model, occupancy map, and planned paths. For the purpose of this project, RViz2 is chosen over Gazebo for visualization because it is more lightweight and the scope of this project does not require a full physics simulation. Gazebo will be consistently slower in launching and running episodes, which is not ideal for iterative tests and tuning of the RL algorithm. RViz2 provides sufficient visualization capabilities for the 2D occupancy grid and planned paths while allowing for faster iteration cycles. To further speed up iterative development, launch files are configured to bring up all required nodes: the robot state publisher, static TF broadcasters, map server, life cycle manager, and RViz2, making it straightforward to launch the full environment with a single command.

B. Performance Evaluation Criteria

In order to assess the impacts of incorporating RL within the local path planning process, relevant path quality metrics were identified throughout our literature review for evaluating path planner performance. These path quality metrics can be summarized by the following criteria:

- Safety (clearance from obstacles)
- Accuracy (success rate in reaching the goal, amount of deviation from global path plan, etc.)
- Efficiency (total time and distance to goal)
- Smoothness (heading angle variance, velocity variance, etc.)

The path quality criteria listed above ultimately served as the foundation for the performance evaluation of the RL-augmented planner against the baseline planner with standard parameters. Additionally, since it was decided that these criteria would be tied directly to performance; these criteria were also worked directly into our reward function design in order to shape training for the RL agent.

C. RL Architecture

The RL component of the project is a trainable agent that tunes planner parameters directly through the Nav2 stack. Independent from the specific path planning algorithm and RL learning algorithm, two design options were initially explored for overall architecture of the states and actions: a one-time parameter tune, or a dynamic parameter tune. It was decided that the implementation and expected results of a dynamic RL application would be favorable and feasible within the scope of this project. With this approach, the RL agent would be responsible for dynamically tuning the local planner's parameters throughout the navigation process; where the agent's state (observation) vector at each time step includes information about what the robot can perceive (e.g. current pose, velocity, distance from the goal, obstacle cost of the surrounding environment, etc.), and the agent's action vector is responsible for adjusting the planner parameters (via a continuous action space) at each timestep. To support frequent observations and actions throughout the navigation process, the reward function is designed to encourage efficient, smooth, and safe navigation by providing positive rewards for progress towards the goal and negative rewards for collisions, long travel times, and time spent in close proximity to obstacles. The goal of these rewards is to encourage the agent to learn how to guide the robot through complex environments as smoothly and quickly as possible.

1) Training Framework: Training is driven by the Stable-Baselines3 (SB3) library, which interfaces with the simulation through a custom Gymnasium environment. SB3 is widely used for training RL agents and provides a variety of algorithms and utilities for training and evaluation, but one challenge that we had to consider is that it is designed for synchronous environments. This project heavily relies on ROS2, which is fundamentally an asynchronous setup with multiple nodes communicating through topics and services.

The main training loop needs to block and wait for new observations and rewards from the ROS2 environment, but also needs to be able to control the asynchronous ROS2 nodes and services to set up and run each episode. To bridge this gap, a state machine-like architecture was implemented to allow the main training loop to operate synchronously with each episode. This was achieved by setting up a thread queue to put the main training loop in charge of the episode flow, where the loop listens to the metrics from a ROS2 topic, and then issues service calls to control the episode flow (start episode, switch planners, update parameters) based on those metrics.

To evaluate which reinforcement learning strategy would be most effective for dynamically tuning planner parameters and maximizing the reward function, two RL design approaches were explored in parallel using two different RL algorithms: PPO and Soft Actor-Critic (SAC). Each offers distinct advantages in continuous-control settings, and comparing them directly allows analysis of their computational costs, sample efficiency, and stability characteristics within the constraints of a ROS2-based navigation environment. By training both agents on the same environment, a determination could be made on whether one algorithm demonstrated more favorable results in a simple 2D navigation environment.

PPO is an on-policy actor-critic RL algorithm that updates its policy using batches of freshly collected trajectories (rollouts). Its defining feature is a clipped objective that restricts how far the policy is allowed to move during each update, which helps maintain stability during training. PPO is widely used because it strikes a balance between implementation simplicity, robustness, and reliable convergence across many continuous-control tasks. Conceptually, PPO learns by repeatedly sampling new experience, computing policy gradients from that experience, and applying carefully constrained updates that prevent the policy from changing too abruptly.

SAC [11] is an off-policy actor–critic RL algorithm that learns from a replay buffer of past transitions rather than relying solely on newly collected data. SAC optimizes expected return and policy entropy, encouraging the agent to maintain broad exploration throughout training. This entropy-regularized objective makes SAC highly sample-efficient and well suited for environments where each interaction step is expensive to generate. At a high level, SAC continuously improves both a stochastic policy and a pair of Q-value critics using stored experience, enabling stable learning even in high-dimensional continuous action spaces.

While the specific reward functions ultimately deviated between the two different RL design approaches, the overall design of the reward functions were the same. The design involved a hybrid reward function which included terms for both dense and sparse rewards based on the relevant path quality evaluation criteria. The dense rewards for both approaches (calculated on a per time step basis) were shaped by progress made toward the goal (Efficiency), commanded angular velocities (Smoothness), and proximity to obstacles (Clearance). The sparse rewards for both approaches involved issuing large positive bonuses for reaching the goal (Accuracy), and penalties for collisions (Safety).

2) Evaluation Framework: Since the path quality evaluation criteria were built directly into the overall reward function design, the respective reward functions were used to evaluate the "score" of both the RL-augmented and baseline planners. Since the reward is already computed for each step through the custom gymnasium environment, the performance evaluation was accomplished by having both the RL-augmented planner (operating with a frozen policy) and the baseline planner step through the custom gymnasium environment over the same predefined set of episodes.

3) Training and Evaluation Tools: To support debugging during both training and evaluation, a live parameter visualization tool was developed. It subscribes to a topic containing the Nav2 parameters being tuned and renders each tunable parameter as a horizontal bar dial in a real-time matplotlib window, updating at 10 Hz. Each dial shows the full valid range for that parameter, a tick mark at the preset baseline value, and a colored fill indicating the current value: blue if the agent has pushed the parameter above baseline and orange if below. An example of the visualization is shown in Figure 1.

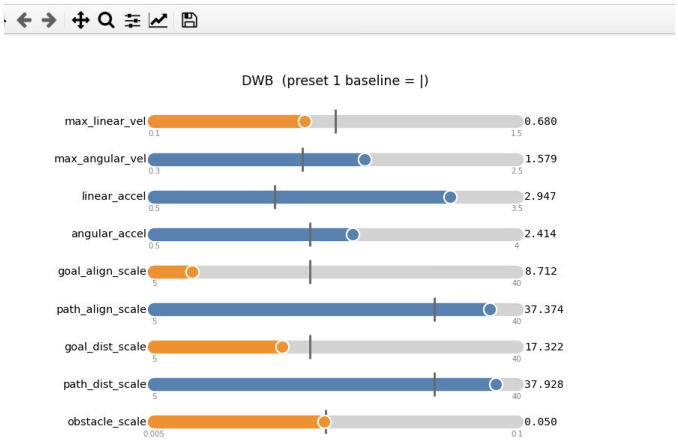

Fig. 1: Live parameter dial visualization during a training run. Each bar shows the current value relative to the preset baseline

The tool proved useful for catching two failure modes that were not obvious from reward curves alone. In one case, the agent collapsed all parameters toward one extreme of their range and kept them there, which the dials made immediately apparent. In another case, the parameters were not changing at all between steps, which pointed to a bug in how actions were being applied rather than a training issue. Having a direct view of what the agent was actually doing to the planner at each step made these problems straightforward to diagnose.

On top of the parameter visualizer, SB3 included wrappers for monitoring the training process in real time through TensorBoard. Since training would typically take several hours, this tool was absolutely necessary for stopping poor training iterations early due to bad sets of hyperparameter configurations (e.g. learning rates, batch sizes, etc.). In addition to allowing

for stepping through training hyperparameter configurations more quickly; TensorBoard also provided history for tracking trends across the different hyperparameter configuration and other design changes (see Figure 2 below).

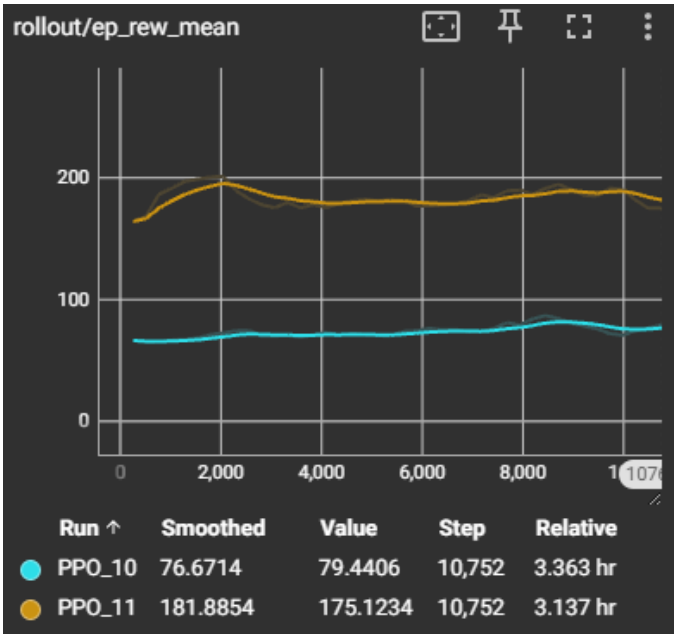

Fig. 2: TensorBoard plot from PPO training before (blue–PPO_10) and after (orange–PPO_11) resolving a bug where episodes with aborted Nav2 path plans were rewarded with the same bonus as episodes which successfully reached the goal. Since drastically different results were given the same reward across episodes, the agent was confused about which actions were "better" and learning was ultimately stifled.

D. General Code Architecture

1) Planner control and Episode Runner: These two ROS2 nodes (planner_controller.py and episode_runner.py) work in concert and act as the backbone for the episodic training loop. The planner_controller exposes three services: one to switch the active planner, one to apply a named preset, and one used by the RL bridge to push continuous parameter updates at each training step. When any of these are called, the node translates the logical parameter names into the ROS parameter names expected by Nav2 and forwards them directly to the controller server. It also keeps the velocity smoother's speed caps in sync and publishes a latched snapshot of the current parameter values for real-time diagnostics (see Figure 3).

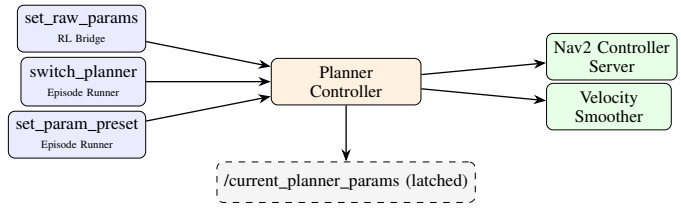

Fig. 3: Planner Controller data flow. Services on the left accept requests from the RL bridge (continuous parameter updates) and the episode runner (planner/preset switching). Translated ROS parameters are pushed to the Nav2 controller server and velocity smoother; a latched snapshot is published for diagnostics.

The episode runner drives the training loop as a state machine. In benchmark mode it automatically cycles through planner and preset configurations, running each episode in sequence. In RL mode it waits after each episode until the training bridge calls `/start_episode`, at which point it teleports the robot to the start pose, waits for the simulation to settle, resets the metrics tracker, and sends the navigation goal to Nav2. When the goal completes (or is cancelled on a collision), it collects the final episode metrics and publishes them for the RL bridge to consume as the episode's terminal signal. The full state machine is shown in Figure 4.

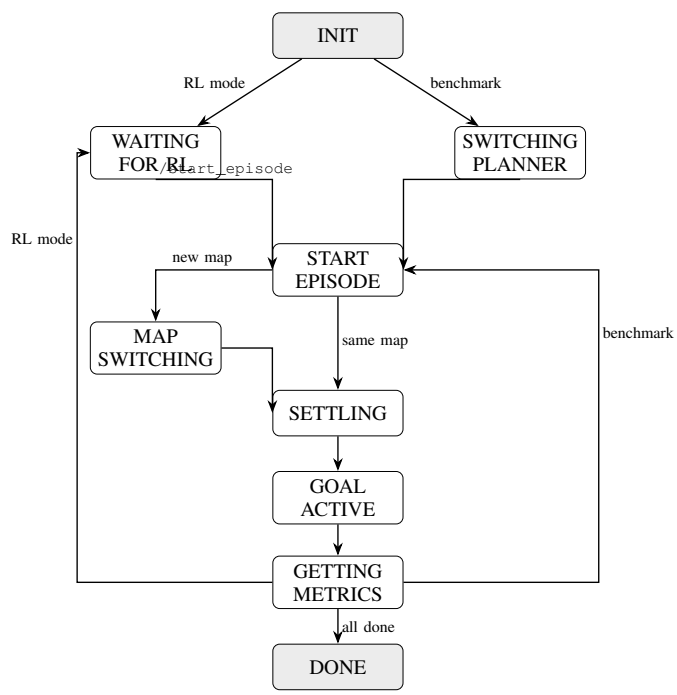

Fig. 4: Episode runner state machine. In RL mode the node waits for `/start_episode` service calls from the training bridge; in benchmark mode it advances automatically. After collecting end-of-episode metrics, it returns to WAITING FOR RL or advances to the next episode or planner configuration.

III. Experimentation

A. Standard Path Planners

In order to determine the feasibility of constantly tuning planner parameters, the system was first tested by manually switching between two parameter sets for both MPPI and DWB. While DWB was able to handle the parameter changes without issue, MPPI struggled to keep moving when parameters were adjusted. This is due to the fact that MPPI relies on a sampling-based approach to generate its path, and when parameters are changed, the samples that were previously valid may no longer be valid, causing the planner to struggle to find a new path. Every time the parameter sets were changed, the robot would come to a stop, and then the planner would slowly resume planning motion. This is not ideal, and dramatically slows down the navigation process, defeating the purpose of this project. While it may be possible to mitigate this issue, the team decided to concentrate focus on the RL agent, and DWB was ultimately chosen to be the only planner that the RL agent would be trained on. DWB is a more traditional local planner that relies on a costmap-based approach, and it is able to handle parameter changes more gracefully than MPPI. This is because DWB generates its path based on the costmap, which is updated in real-time as the robot moves through the environment. As a result, when parameters are changed, DWB can quickly adapt to the new costmap and continue planning without significant disruption.

B. RL-Trained Planner: SAC Policy Agent

1) DWB Parameters: Four DWB parameters were selected for tuning to keep the action space small and learning tractable. Maximum linear velocity controls how fast the robot can move and has a direct effect on efficiency. Obstacle scale controls how aggressively DWB avoids obstacles when scoring candidate trajectories, governing the tradeoff between clearance and speed.

The other two are meta-parameters: path weight and goal weight. Rather than tuning DWB's four critic scales independently, each meta-parameter scales a correlated pair together: path weight adjusts both path-following critics simultaneously, and goal weight adjusts both goal-seeking critics. This keeps the action space small without giving up meaningful control, since the two critics in each pair always push in the same direction anyway.

2) State and Action Space Design: The agent observes 12 values at each timestep: eight navigation state values and four values representing the agent's own currently active parameter settings. The navigation state includes three path-cost lookahead values sampled from the Nav2 costmap along the upcoming plan (covering roughly 1 m, 3 m, and 6 m ahead), distance and heading error to the goal, linear velocity, angular velocity, and lateral deviation from the global plan. Raw LiDAR was excluded because the costmap lookahead already captures obstacle proximity in a form the planner directly acts on, and a smaller observation keeps learning more tractable. The agent's own smoothed parameter values are appended to the observation so the policy can see what settings are currently active in the planner.

The action space is four continuous values, one per tunable parameter. The raw policy output is smoothed with an exponential moving average before being decoded into physical values and applied to Nav2, which prevents rapid swings from disrupting DWB mid-episode. Obstacle scale is decoded on a log scale since its useful range spans two orders of magnitude; the other three parameters are decoded linearly.

3) Reward Function Design and Tuning: The reward function was adjusted several times during training to address observed failure modes. The core signal is a progress reward for reducing distance to the goal each timestep, with smaller penalties on path deviation and angular velocity to encourage smooth, on-plan behavior. A graduated proximity penalty also activates when the planned path ahead enters an inflated obstacle region, growing as the path gets closer to the obstacle.

Two terms were added during tuning. A direct collision penalty was added after observing that the proximity penalty alone was not a strong enough deterrent when the robot actually entered an obstacle cell. A slow-pace penalty was added to address cases where the agent learned to creep slowly through open areas, it only fires when the robot is far from the goal and well clear of obstacles, so it does not punish legitimate slowdowns near corners or as the robot approaches the goal.

The original flat time penalty per timestep was replaced with a per-goal time-delta terminal reward that compares the agent's completion time against a pre-collected baseline time for that specific goal. This prevents longer routes from automatically producing larger penalties just because they take more time. Baseline times are gathered by running the fixed preset-1 configuration through all goal configurations before training begins. At episode end, a goal bonus of $+100$ is awarded for success and a failure penalty of -50 is applied otherwise. A collision also immediately ends the episode with the failure penalty rather than letting navigation continue, since recovery behavior from a collision tends to be erratic and unhelpful for training.

C. RL-Trained Planner:PPO Policy Agent

1) State and Action Space Design: The PPO agent observation space was designed to include information about environment observations, robot location, and anticipated heading. For what the robot sees, the state vector includes space for a predefined set of N downsampled lidar ranges and bearings. However, instead of downsampling the lidar measurements in fixed increments around the robot, the lidar rays were divided into N segments; where the minimum range measurements and associated bearing angles are extracted from each individual segment. This ensures that lidar measurements associated with the closest features surrounding the robot make it into the state vector and are not missed or sampled out. The lidar segment design is illustrated in Figure 5.

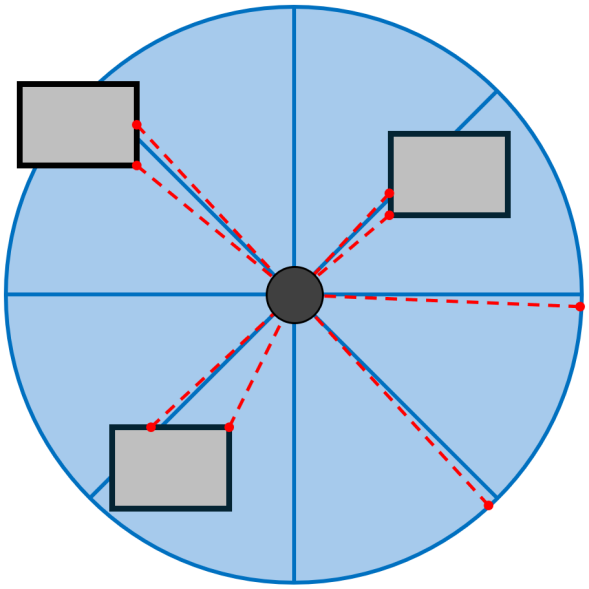

Fig. 5: Lidar segment design illustration. As part of this illustration, the turtlebot (dark grey center circle) is located between 3 obstacles (light grey rectangles). The turtlebot's span of lidar scans (light blue circle) are broken up into 8 segments (dark blue slices). Within each lidar segment, the lidar scan with the minimum range (red dashed lines) would be inserted into the state vector. For segments that don't have a distinct minimum lidar range, the first clockwise lidar scan would be inserted into the state vector.

For where the robot thinks it is, the state vector included the error components for x, y, and yaw. The error components were selected over the global frame components for x, y, and yaw as an attempt to minimize the dimensionality of the state vector; because the agent would either require knowledge of the goal pose, or require knowledge of the map itself, in order to make sense of the global frame components. And from a function approximation perspective, it makes more sense intuitively to determine actions based on how far away the robot is from the goal over where the robot is located within a given map. This is especially true when the same global coordinates represent entirely different states across different maps. For where the robot thinks it's going, the state vector included the elements for the linear and angular velocity components of the robot. Since the Turtlebot that we're trying to control is a differential drive robot, the velocity components can be represented concisely by a single forward velocity term, v, and a single steering velocity term, ω. All together, the state vector took the following form:

$$s = \begin{bmatrix} r_{lidar_0} \\ \theta_{lidar_0} \\ \dots \\ r_{lidar_(N-1)} \\ \theta_{lidar_(N-1)} \\ x_{err} \\ y_{err} \\ \theta_{err} \\ v \\ \omega \end{bmatrix}$$

where:

- s is the state (observation) vector.
- N is the number of lidar segments.
- $r_{lidar_(N-1)}$ is the range of the shortest lidar return in segment $(N-1)$.
- $\theta_{lidar_(N-1)}$ is the bearing (with respect to the robot) of the shortest lidar return in segment $(N-1)$.
- x_{err} is the x-component of the error between the robot's current pose and the goal pose.
- y_{err} is the y-component of the error between the robot's current pose and the goal pose.
- θ_{err} is the yaw-component of the error between the robot's current pose and the goal pose.
- v is the linear (forward) velocity of the robot.
- ω is the rotational (steering) velocity of the robot.

While different configurations of the state vector were tested, the final state vector included lidar measurements from 18 segments (N=18); so the final state vector consisted of 41 elements.

The PPO agent action space was specifically designed to configure how fast, and how smooth, the DWB planner should control the robot. Of course, this is all done indirectly by adjusting boundaries and constraints that the DWB planner can work within. For configuring how fast the robot is controlled, the action vector includes the DWB parameters that control both the maximum and minimum velocities (increasing the maximum velocity alone may not guarantee that the robot will speed up without also increasing the minimum velocity). For controlling how smooth the robot is controlled, the action vector includes the DWB parameters which control both the goal seeking and path seeking weights. This indirectly influences the smoothness of the planner because it effectively tells DWB how closely it should follow the immediate waypoints within the global path versus ignoring the global path to a certain degree in favor of reaching the goal faster. All together, the action vector took the following form:

$$a = \begin{bmatrix} v_{max} \\ v_{min} \\ GoalAlign.scale \\ PathAlign.scale \\ GoalDist.scale \\ PathDist.scale \end{bmatrix}$$

where:

- v_{max} is the DWB planner's "max_vel_x" parameter (velocity ceiling).
- v_{min} is the DWB planner's "min_speed_xy" parameter (velocity floor).
- $GoalAlign.scale$ is the DWB planner's "GoalAlign.scale" parameter (scores a trajectory based on how well aligned the trajectory is with the goal pose).
- $PathAlign.scale$ is the DWB planner's "PathAlign.scale" parameter (scores a trajectory based on how well it is aligned to the path provided by the global planner).
- $GoalDist.scale$ is the DWB planner's "GoalDist.scale" parameter (scores a trajectory based on how close the trajectory gets the robot to the goal pose).
- $PathDist.scale$ is the DWB planner's "PathDist.scale" parameter (scores a trajectory based on how close the trajectory gets the robot to the trajectory of the global planner).

With the state and action spaces of the PPO agent defined, the feedback between the state and action mapping needed to be established. While the reward function plays a pivotal role in providing that feedback for the agent to learn from; additional control was needed to account for the asynchronous nature of ROS2, along with the Nav2 stack's sensitivity to large rapid changes. While the custom gymnasium environment was responsible for transmitting the action to Nav2, the underlying ROS2 environment had potential to propagate the action to the Nav2 stack and publish an observation for the next state simultaneously. Without intervention, there would be no way to guarantee the operating assumption of RL environments where the next state is produced as a result of the recently applied action. And even though there is a way to guarantee that the next state observation is published after the Nav2 action is applied (via ROS2 callback groups); the immediate next state observation that follows the applied Nav2 action isn't guaranteed to be "useful" information for the agent since the impact of the DWB planner adjustments may not have had time to be fully realized. So in order to provide enough time for the planner's current configuration to produce a meaningful next state, the PPO agent's custom gymnasium environment was programmed to wait for a pre-specified amount of time before accepting a newly published observation from ROS2. The final implementation of the PPO agent's environment ended up using a 1.0 second wait time, but this parameter could ultimately be treated as a hyperparameter and further optimized.

2) Reward Function Design: The hybrid reward function for the PPO could be summarized as follows:

$$R = r_{terminal} + r_{step} + r_{progress} + r_{smoothness} + r_{proximity}$$

TABLE I: PPO Agent Reward Function Constants

Parameter	Value	Description
R_{GOAL}	+100	Sparse bonus for the robot successfully reaching the goal.
R_{COLL}	-100	Sparse penalty for when the robot experiences a collision.
R_{STEP}	-0.01	Dense penalty for steps through the environment meant to encourage faster paths toward the goal.
R_{PROG}	+10.0	Dense reward scale meant to encourage making progress towards the goal and discourage further distancing from the goal.
R_{SMOOTH}	-0.01	Dense reward scale meant to discourage unnecessary or jerky rotation.
R_{PROX}	-1.0	Dense reward scale meant to penalize the robot's proximity to environment features within the minimum range threshold, $r_{threshold}$.
$r_{threshold}$	1.0	The minimum range threshold (in meters) which activates the proximity penalty. When lidar measurements fall below this threshold, the penalty is determined by scaling R_{PROX} by the difference between the threshold and the minimum lidar range measurement.

where:

$$r_{terminal} = \begin{cases} R_{GOAL} & \text{if goal reached} \\ R_{COLL} & \text{if collision occurred} \\ 0 & \text{otherwise} \end{cases}$$

$$r_{step} = R_{STEP}$$

$$r_{progress} = R_{PROG} * (xy_error_{last} - xy_error_{current})$$

$$r_{smoothness} = R_{SMOOTH} * |\omega|$$

$$r_{proximity} = R_{PROX} * (r_{threshold} - r_{min_lidar}))$$

and the values of the final PPO agent implementation are contained in Table I.

IV. Results

A. RL-Trained Planner: SAC Policy Agent

Table II summarizes the results of a 300-episode evaluation comparing the fixed DWB preset 1 baseline against the trained SAC policy. Figure 6 shows the corresponding time-to-goal distributions for successful episodes.

TABLE II: 300-episode evaluation: DWB preset 1 baseline vs. SAC policy.

Metric	Baseline	Policy	Delta
Episodes	300	300	
Goals reached	288	290	+2
Success rate	96.0%	96.7%	+0.7 pp
Mean time (successes)	37.9 s	38.7 s	+0.79 s
Median time (successes)	36.0 s	36.5 s	+0.50 s
Std time (successes)	12.38 s	12.35 s	−0.03 s

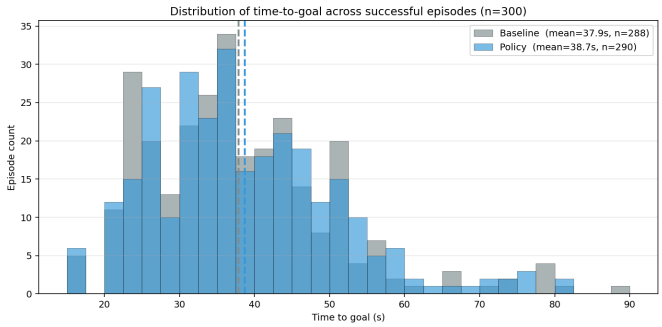

Fig. 6: Distribution of time-to-goal across all successful episodes (n=300). Dashed lines mark the mean for each condition. The two distributions are largely overlapping, indicating the SAC policy matches the baseline in navigation speed while achieving a marginally higher success rate.

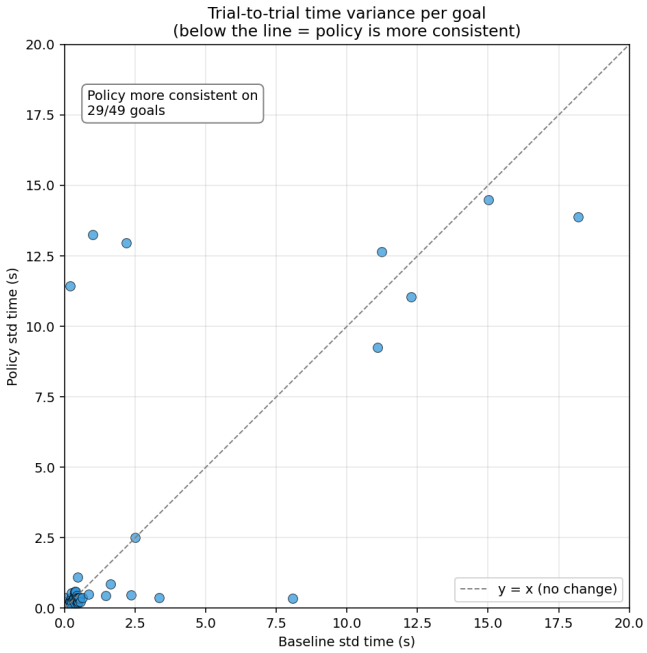

Fig. 7: Per-goal trial-to-trial time standard deviation: baseline vs. SAC policy. Each point represents one of 49 goal configurations. Points below the diagonal indicate the policy is more consistent than the baseline on that goal. The policy achieves lower variance on 29 of 49 goals, a pattern obscured by the near-identical aggregate standard deviations in Table II.

B. RL-Trained Planner: PPO Policy Agent

The PPO agent was trained for 20,000 steps with various start and goal configurations spread across 3 different maps. The training performance is highlighted in Figures 8 through 10.

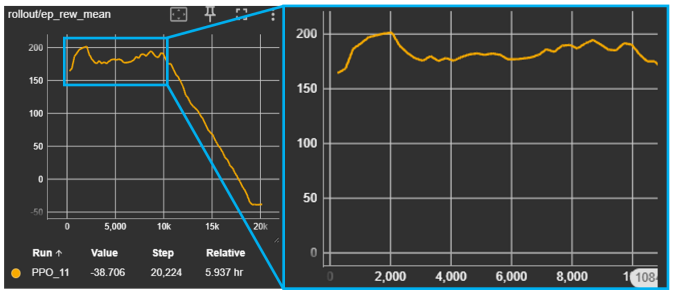

Fig. 8: TensorBoard plot of the mean (rolling average over 100 episodes) episode rewards over the entire span of training (left). Since the reward began to significantly diverge sometime after 10,000 steps, the 10,000 step checkpoint was used during evaluation; so the effective training curve is clipped from the entire span and highlighted in blue (right). Over the 10,000 steps, the reward oscillates but generally trends upward, indicating that the PPO policy is learning improved configurations over the baseline.

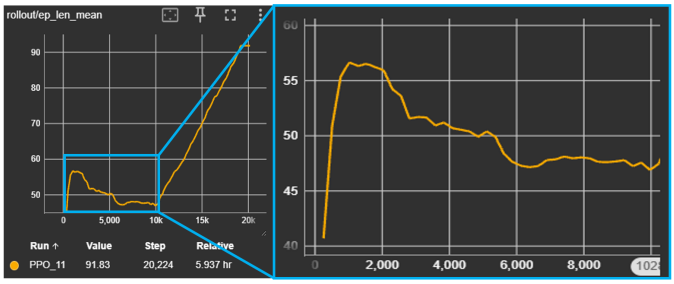

Fig. 9: TensorBoard plot of the mean (rolling average over 100 episodes) episode length (in terms of steps) over the entire span of training (left). Divergence after 10,000 steps aligns with Figure 8, so the effective training curve is also clipped and highlighted in blue here (right). Over the 10,000 steps, the total number of steps per episode noticeably trends downward, indicating that the PPO policy is specifically learning more efficient paths to the goal over the baseline.

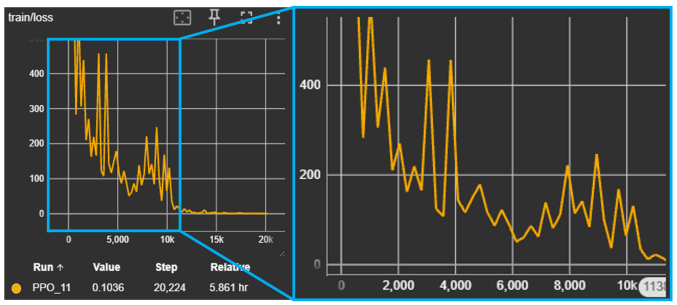

Fig. 10: TensorBoard plot of the total loss over the entire span of training (left). This encompasses all components of loss, including the actor's policy gradient loss and the critic's value loss; and is generally a good indicator of how effectively the agent is learning to maximize rewards while maintaining stability. However, when considering the divergence after 10,000 steps captured in Figures 8 and 9, the curve over the entire span of training can be a bit misleading. The total loss may appear to significantly improve after 10,000 episodes, but this is because the total loss is dominated by the critic network's value loss. After 10,000 steps, the policy diverges to making poor decisions, and the critic begins to accurately assess the low reward values of the chosen actions. This is why the reward curve can appear to degrade while the loss curve appears to improve. Regardless, the clipped loss over the 10,000 steps (right) does noticeably trend downward, indicating that the PPO policy does learn to accurately map actions to their expected rewards and specifically take actions that maximize the reward.

It's clear from the training performance that the PPO agent's policy matured over the span of 10,000 episodes; after which it began to degrade. So instead of using the final state saved at the end of training, the PPO agent's 10,000 step checkpoint was used during evaluation. Evaluation was performed for the fixed DWB preset baseline and the trained PPO agent over the same fixed set of 200 episodes. The results are summarized in Table III.

TABLE III: Evaluation of DWB preset vs. PPO agent

Metric	Baseline	PPO Agent	Delta
Episodes	200	200	
Success rate	86.0%	91.0%	+5%
Collision rate	3.0%	0.5%	−2.5%
Timeout rate	4.0%	4.0%	
Stuck rate	7.0%	4.5%	−2.5%
Mean reward	183.16	196.73	+13.57
Mean reward (successes)	219.64	219.86	+0.22
Std reward	97.33	81.38	−15.95
Std reward (successes)	34.86	34.55	−0.31
Mean steps	42.62	40.18	−2.44

V. Discussion

A. SAC Policy Agent

Overall, the SAC policy performed comparably to the fixed DWB baseline, offering marginally better success rates at the cost of a minimal increase in time to goal. The policy's ability to adapt planner parameters based on the robot's and local path's surroundings seems to provide a noticeable boost in performance on the more difficult episodes.

1) Overall Results: On 300 evaluation episodes, the SAC policy achieved a 96.7% success rate, slightly outperforming the 96.0% success rate of the fixed DWB baseline. The mean time to reach the goal across successful episodes was 38.7 s for the SAC policy, compared to 37.9 s for the baseline, indicating a negligible increase in navigation time. The distributions of time-to-goal were largely overlapping between conditions, suggesting that the SAC policy matches the baseline in navigation speed while achieving a marginally higher success rate.

2) Per-Goal Adaptation Performance: The policy offers a small performance boost on the baseline DWB planner's success rate, most of these gains come from the more difficult episodes with goals that are harder to reach, where the baseline planner is unable to reliably complete the episode. In these cases the policy's ability to adapt the planner parameters in response to the specific environment conditions around the local plan seems to provide a boost in performance. However, this is not always the case. In certain episodes where the baseline is sometimes able to complete the episode (still with a very low success rate), the policy seems to struggle more than the baseline. This could likely be due to the fact that the policy was trained on only one map and set of goals, but the tests were to determine whether the policy could generalize to new maps and goals. So while the policy is able to provide a boost in performance on some of the more difficult episodes, it also struggles to generalize to certain episodes that the baseline can complete.

3) Time to Goal Performance: While the policy was able to achieve a slightly higher success rate than the baseline, it did so with a slightly higher mean time to goal across successful episodes. This could be due to the fact that the policy is adjusting the planner parameters in response to the local environment conditions, which may lead to more cautious behavior in certain situations. For example, if the policy detects that the robot is approaching an obstacle, it may adjust the parameters to prioritize safety over speed, resulting in a longer time to reach the goal. However, since the distributions of time-to-goal were largely overlapping between conditions, this increase in time does not seem to be a significant tradeoff for the boost in success rate.

4) Safety Performance: One of the concerns in this project was that the policy would learn to exploit the action space in a way that would lead to unsafe behaviors being rewarded for the sake of time efficiency. For example, the policy could learn to increase the maximum velocity and lower the obstacle avoidance weights to speed up navigation, but this could cause the robot to drive aggressively and collide with obstacles. However, the graduated proximity to obstacle cost penalty in the reward function seems to have successfully prevented this type of behavior. The policy seems to have learned to match the baseline's performance in terms of safety, as measured by the number of collisions across all episodes along with the average clearance and time spent in obstacle inflation.

5) Performance Consistency: The mean time for episode completion was very similar between the policy and the baseline. However, the policy seems to reduce the variance on 29 of the 49 goal configurations. This suggests that the policy is usually able to react the same way to the same goal and surroundings across the path. The variance between some of the harder goals went from 3.25 seconds in the baseline to 0.6 seconds, and from 1.89 seconds to 0.002 seconds. This indicates that the policy is able to provide a more consistent navigation performance across episodes, which could be beneficial for real-world applications where predictability is important. DWB likely has some variance when navigating through tight spaces due to the fact that it's somewhat aggressive behavior will allow the robot to get very close to obstacles, and small differences in the local environment will determine how long the robot will get stuck in high cost regions of the costmap.

6) Limitations: The policy was only trained on one map and set of goals, so its ability to generalize to new maps and goals is limited. The team made a choice to do it this way to determine whether the policy could be trained to generalize to various new maps and goals. However, the weakness in this approach is that the policy may struggle to adapt to certain maps' unique features that aren't well represented in the training map.

One other limitation is that the simulation environment does not factor in physics or dynamic obstacles. While this simplified the environment setup for the project, it also means that the policy may not be able to handle certain real world factors, such as wheel slip or moving obstacles.

The reward function design also has a potential issue to be addressed in future work. The reward function is set up for the policy to learn the most in scenarios where the standard DWB planner struggles. This is because the reward function rewards the policy for beating the time of the baseline DWB planner in a given episode, so the policy has more to gain by improving performance on episodes where the baseline DWB planner gets the robot stuck and takes a long time to free itself. While this helps the robot achieve a higher success rate on more difficult episodes, it could be the cause of the policy's average episode completion time being very slightly higher than the baseline's.

B. PPO Policy Agent

Overall, the PPO policy performed noticeably better than the baseline policy in terms of success rate and efficiency toward the goal. When only considering episodes where the turtlebot successfully reaches the goal, the PPO policy performed marginally better than the baseline in terms of total reward.

1) Safety Performance: Of the 5% increase in success rate from the PPO policy (10 episodes), half of those episodes (5) came directly from avoiding collisions. This is even more impressive when considering the fact that safety performance seemed to improve without sacrificing efficiency, since the mean number of steps to reach the goal decreased with the PPO policy. While the team wasn't able to build out a more granular breakdown of the reward function components during evaluation, the notion that safety improved without sacrificing efficiency is further corroborated by the marginal improvement in rewards per episode. Since the penalty for proximity to obstacles, R_{PROX}, in Table I is significantly larger than the step penalty, R_{STEP}, the reward improvement wouldn't occur if the PPO policy spent more time with closer proximity to obstacles than the baseline.

2) Implications of Episode Failure Modes: Apart from collisions, the other failure modes for the PPO agent's environment were broken out into "timeouts" and "getting stuck". On the surface, both failure modes seem similar, but the distinction comes from the source of the terminal state: timeouts occur when the maximum number of steps within a given episode is exceeded; and the "stuck" state is assigned when Nav2 aborts the navigation plan. Despite neither failure mode resulting in a direct penalty or bonus in the PPO agent's reward function, the PPO agent effectively reduced the number of plans aborted by Nav2 while remaining ineffective at preventing timeouts. This implies that the PPO policy was able to improve Nav2 stability in addition to improving the safety and efficiency of the navigation performance.

3) Limitations: In contrast to the SAC training and evaluation, the PPO policy was trained across a variety of maps that were re-used as part of the evaluation process. So while the PPO policy offered improvements over the baselines, not much can be said in terms of generalization. Ideally, machine learning workflows would involve a wide variety of training scenarios along with a small subset of test scenarios that aren't specifically trained on. However, the team's sample size was limited due to the manual process of configuring numerous start and goal poses that are valid (within grid boundaries and not unreachable) for each map.

Additionally, as discussed as part of the SAC Limitations, the simulation environment does not factor in physics or dynamic obstacles, so the PPO policy that was evaluate may not extend as successfully to real world factors (e.g. wheel slip or moving obstacles).

C. Future Work

Looking to future applications of RL-enhanced path planning along side the work that has already been investigated in this report, one of the next logical steps is to implement a wider range of path planning algorithms such as Dijkstra's, A*, rapidly-exploring random tree (RRT), breadth-first or depth-first search (BFS, DFS) or others in order to see if there would be benefits of augmenting them with an on-policy or off-policy RL agent. The addition of other path planners would also encourage the development of a planner switch. This is something that was originally discussed as a goal for this project, but it was decided to explore varied RL agents instead. Given that there is now a base of two different RL approaches, the simulation would need to be updated to account for the correct hyperparameters that each planner would need, but with the rest of the structure being already built as adaptable around that, the major changes would be in the navigator config yaml file(s).

Future work should also revisit the reward function design. As discussed above, one of the current reward functions unintentionally rewards the agent more for outperforming the baseline on difficult episodes than on simple ones, which distorts the agent's learning. Multi-objective reward shaping, adaptive weighting, or learned reward models could produce more stable and reliable behavior while ensuring that improvements are meaningful across all episode types.

With the addition of more path planners, a switching option could be implemented wherein the agent could select the best path planner for a crowded or more dynamic section of a map, utilize it, then switch to a more lightweight one later on. This would require a much more in-depth and expanded hyperparameter base as touched on above, but could prove beneficial if overall computational load could be reduced. Another important extension is incorporating physics-accurate simulation and dynamic obstacles. The current environment does not model robot or load inertia, wheel slip, or moving obstacles, which limits the realism of the agent behavior. Integrating a physics simulator such as Gazebo would allow the agent to learn parameter-tuning strategies that account for real-world motion constraints on top of dynamic interactions with pedestrians, forklifts, or other robots.

Lastly, as project development progressed for both of the RL architecture approaches, the set of maps that were intended for training and evaluation of the agents was not scaled appropriately. This was primarily due to time constraints for the project, so in spirit of casting a wide net, each approach took a different strategy to training and evaluation. The first step towards a more realistic result would be unifying training for both agents on a large set of different heterogeneous maps and testing the agents on a distinct smaller set of maps. The maps should vary in size, complexity, aisle size, as well as continue to use the randomly generated start and goal points / poses. In that same line of thinking, this projects' maps were relatively low-clutter, 2D, static environments; therefore putting these same agents in crowded and / or dynamic traffic environments would be a step towards realistic deployment. Warehouses are full of people, forklifts, and clutter so testing in a sterile environment (as was done in this simulation) is a great start, but does not accurately describe real-world implementation. Additionally, with dynamic obstacles and multiple maps, it may become more clear if either the PPO or SAC agents find more of an advantage over standard planners and extend the marginally positive results seen in this project.

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APPENDIX

A. Source Code

Due to the extensive repository and length of the simulation, all the code for this project can be found in a Git repository here: [https://github.com/SushiRaj99/rbe-capstone].